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Languages & Libraries

From:

Testbench + Design

SystemVerilog/Verilog

To:

UVM / OVM (http://eda-playground.readthedocs.org/en/latest/intro.html#libraries-methodologies)

None

Other Libraries (http://eda-playground.readthedocs.org/en/latest/intro.html#libraries-methodologies)

None

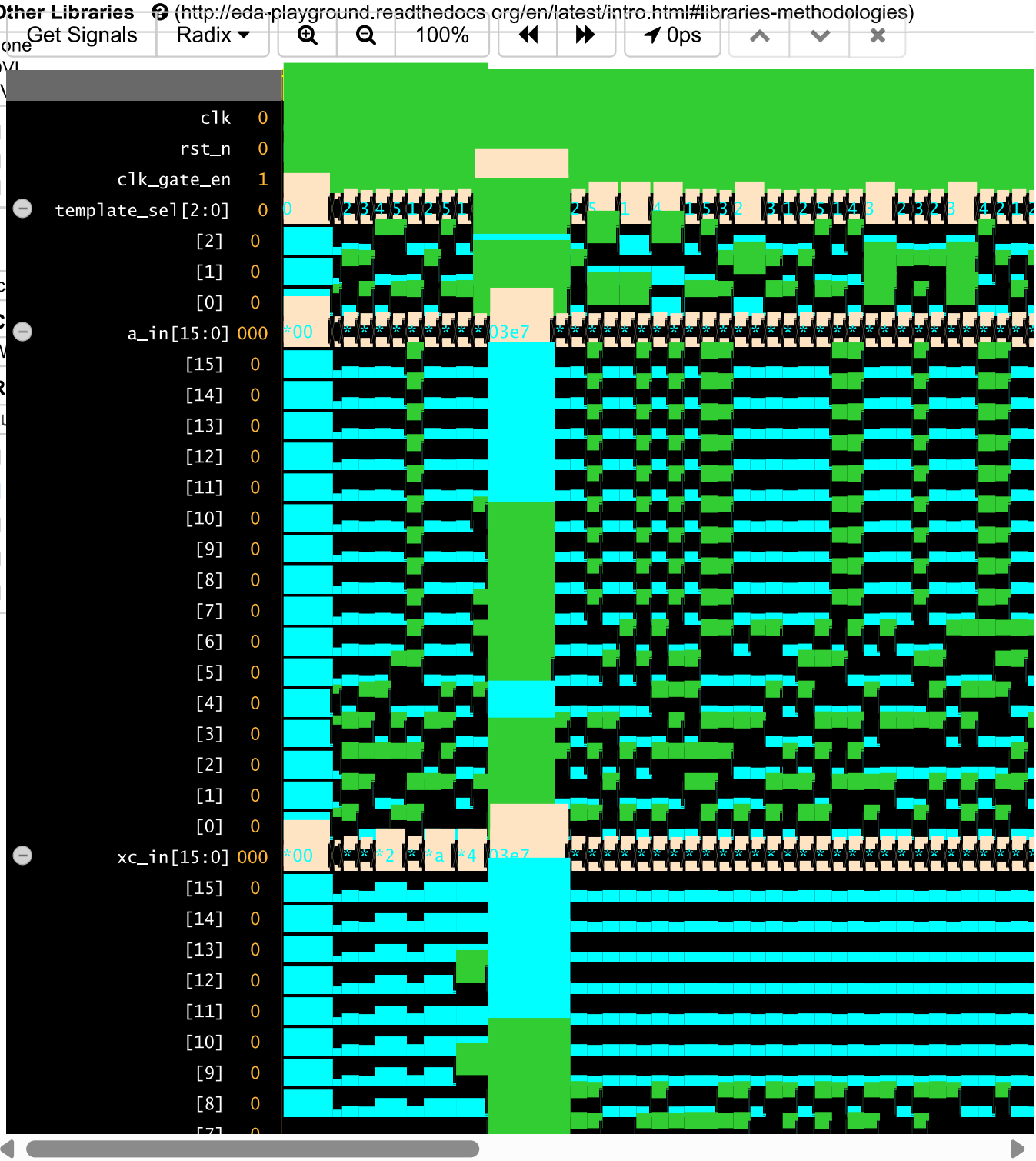
OVI

SV

Instructions: x

Click on an edge to display a cursor. **CTRL-click** on another edge to display a second cursor.

Note: To revert to EPWave opening in a new browser window, set that option on your profile page.



searchString=&language=Specman+e+%2B+SV%2F+Verilog&simulator=&methodologies=&curated=true)

Python + Verilog (/playgrounds?searchString=&language=Python+%2B+SV%2FVerilog&simulator=&methodologies=&curated=true)
Python + VHDL (/playgrounds?searchString=&language=Python+%2B+VHDL&simulator=&methodologies=&curated=true)
Python Only (/playgrounds?searchString=&language=Python+2.7+only&simulator=&methodologies=&curated=true)
C++/SystemC (/playgrounds?searchString=&language=C%2B%2B+only&simulator=&methodologies=&curated=true)

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testbench.sv

```
// ============================================================================
// Comprehensive Self-Checking Verilog Testbench for FCU Datapath
// Tests:
//   1. Templates T1 - T5 with positive operands
//   2. Templates T1 - T5 with negative operands
//   3. Boundary & Corner Cases (0x7FFF, 0x8000, 0x0000, 0xFFFF)
//   4. Glitch-Free Dual-Rail Integrated Clock Gating (ICG) hold/resume
//   5. 100 Randomized equivalence trials with auto-assertion
// ============================================================================
`timescale 1ns / 1ps

module testbench;

    parameter WIDTH = 16;

    reg                     clk;
    reg                     rst_n;
    reg                     clk_gate_en;
    reg  [2:0]              template_sel;
    reg  signed [WIDTH-1:0] a_in;
    reg  [WIDTH-1:0]        xc_in, xs_in;
    reg  [WIDTH-1:0]        yc_in, ys_in;
    reg  [WIDTH-1:0]        kc_in, ks_in;
    wire [WIDTH-1:0]        wc_out, ws_out;
    wire signed [WIDTH-1:0] bin_out;

    // Instantiate FCU
    fcu_top #(
        .WIDTH(WIDTH)
    ) dut (
        .clk(clk),
```

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design.sv

```
// ============================================================================
// 1. Integrated Clock Gating (ICG) Cell
// Negative-level latch prevents clock glitches on the output gated_clk.
// When clk_gate_en is low, the clock is cleanly gated to save dynamic power.
// ============================================================================
`timescale 1ns / 1ps

module icg_cell (
    input  wire clk,
    input  wire clk_gate_en,
    output wire gated_clk
);
    reg latch_en;

    // Negative-level latch captures enable during low clock phase
    always @(clk or clk_gate_en) begin
        if (!clk) begin
            latch_en <= clk_gate_en;
        end
    end

    assign gated_clk = clk & latch_en;

endmodule

// ============================================================================
// 2. 16-bit 4:2 Carry-Save Compressor / Adder
// Fuses additions and subtractions prior to multiplication in constant 4 Tg delay.
// Uses two cascaded Full Adders per bit slice; carries do NOT propagate horizontally.
// cin1: Carry-in to horizontal carry chain
// cin2: Carry-in into bit 0 of the carry output vector (+2 total for dual-rail ~Y)
// ============================================================================
`timescale 1ns / 1ps

```

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Log Share

```
[2026-11-07 18:19:59 UTC] iverilog '-Wall' '-g2012' design.sv testbench.sv  && unbuffer vvp a.out
VCD info: dumpfile dump.vcd opened for output.
======================================================================
   STARTING ADVANCED FCU VERILOG TESTBENCH (GENUINE CSA TREE)
======================================================================

--- SECTION 1: DIRECTED POSITIVE TESTS ---
>> [PASS] T1 Pos: 25 * (100 + 50) + 10 | T1 | Output: 3760 | wc: 0x0004, ws: 0x0eac
>> [PASS] T2 Pos: 15 * (150 - 30) + 5 | T2 | Output: 1805 | wc: 0x4100, ws: 0xc60d
>> [PASS] T3 Pos: 30 * 12 + (60 + 40) | T3 | Output: 460 | wc: 0x10a0, ws: 0xf12c
>> [PASS] T4 Pos: 20 * 10 + (100 - 40) | T4 | Output: 260 | wc: 0x0214, ws: 0xfef0
>> [PASS] T5 Pos: 45 * 16 | T5 | Output: 720 | wc: 0x0000, ws: 0x02d0

--- SECTION 2: SIGNED & NEGATIVE OPERAND TESTS ---
>> [PASS]  Neg Multiplicand: -25 * 50 + 50 | T1 | Output: -1200 | wc: 0x0024, ws: 0xfb2c
>> [PASS] Neg Difference: 10 * (-30) + 100 | T2 | Output: -200 | wc: 0x0088, ws: 0xfeb0
>> [PASS] T5 Neg K: 30 * (-15) | T5 | Output: -450 | wc: 0x0000, ws: 0xfe3e

--- SECTION 3: BOUNDARY & CORNER CASES ---
>> [PASS] Zero Multiplicand: 0 * X + 77 | T1 | Output: 77 | wc: 0x0000, ws: 0x004d
>> [PASS] Zero Multiplier: 1234 * 0 | T5 | Output: 0 | wc: 0x0000, ws: 0x0000

--- SECTION 4: DUAL-RAIL INTEGRATED CLOCK GATING (ICG) ---
Asserting clk_gate_en = 0 (Gating clock)...
>> [PASS] ICG Hold: wc_out (0x0000) and ws_out (0x0000) strictly held across 4 cycles!
```

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```
Re-enabling clk_gate_en = 1 (Resuming clock)...
>> [PASS] ume: Computation resumes cleanly | T5 | Output: 120 | wc: 0x0000, ws: 0x0078

--- EXECUTION 5: 100 RANDOMIZED EQUIVALENCE TESTS ---
>> Completed 100/100 randomized trials.

======================================================================
   FINAL TESTBENCH EXECUTION SUMMARY
======================================================================
   Total Test Vectors Executed : 112
   Total Functional Errors     : 0
   STATUS                      : 100% ALL TESTS PASSED!
   GENUINE CSA TREE MULTIPLIER : FULLY VERIFIED & DEFENDED
======================================================================
testbench.sv:296: $finish called at 1196000 (1ps)
Finding VCD file...
./dump.vcd
 [2026-11-07 18:20:00 UTC] Opening EPWave...
 Done
```

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